

Diego González Sánchez

Alfréd Rényi Institute of Mathematics
Budapest, Reáltanoda utca 13-15, 1053
diegogs (at) renyi (dot) hu
<https://dglez91.github.io/>

Employment

11/2020 – Present	Research fellow , Alfréd Rényi Institute of Mathematics, Hungary
02/2024 – 05/2024	Professor , Budapest Semesters in Mathematics, Hungary

Education

10/2016 – 10/2020	Ph.D. , Autonomous University of Madrid, Spain Thesis: <i>Topics in additive combinatorics and higher order Fourier analysis</i> Supervisor: Prof. Pablo Candela. Defended on 25/09/2020. Final score: Outstanding (<i>sobresaliente</i>) cum laude.
09/2015 – 07/2016	M.A.St. , in Pure Mathematics (Part III), Cambridge University, UK Final score: 73/100, Merit
09/2009 – 06/2015	B.Sc. in Mathematics, Autonomous University of Madrid, Spain. Final score: 9.41/10 outstanding (<i>sobresaliente</i>) B.Sc. in Computer Science, Autonomous University of Madrid, Spain. Final score: 9.41/10 outstanding (<i>sobresaliente</i>)

Visiting periods (at least one month)

05/2019 – 08/2019	Visiting researcher, Alfred Renyi Institute of Mathematics, Hungary
09/2013 – 07/2014	Erasmus student, Technical University of Denmark, Denmark

Papers published or accepted for publication

- [1] *A short proof of an inverse theorem in bounded torsion groups*, (with P. Candela and B. Szegedy). Discrete Mathematics Days (conference proceedings), Alcalá de Henares, July 3-5, (2024). DOI: 10.37536/TYSP5643. Open access: arXiv:2311.13899.
- [2] *On $\mathbb{F}_2^\omega$ -affine-exchangeable probability measures*, (with P. Candela and B. Szegedy). Studia Mathematica, (2024), 79 (1) . Open access: arXiv:2203.08915.
- [3] *On measure-preserving $\mathbb{F}_p^\omega$ systems of order k* , (with P. Candela and B. Szegedy). Accepted for publication at Journal d'Analyse Mathématique. Open access: arXiv:2311.13899.
- [4] *On higher-order Fourier analysis in characteristic p* , (with P. Candela and B. Szegedy). Ergodic Theory and Dynamical Systems, 1-70, (2022). Open access: arXiv:2109.15281.
- [5] *A refinement of Cauchy-Schwarz complexity*, (with P. Candela and B. Szegedy). European Journal of Combinatorics 106, 103592 (2022). Open access: arXiv:2109.05965.
- [6] *Optimal transport with f -divergence regularization and generalized Sinkhorn algorithm*, (with D. Terjék). International Conference on Artificial Intelligence and Statistics (conference proceedings), 5135-5165 (2022). Open access: arXiv:2105.14337.
- [7] *A Refinement of Cauchy-Schwarz Complexity, with Applications*, (with P. Candela and B. Szegedy). Extended Abstracts EuroComb 2021 (conference proceedings), 293-298 (2021). Open access: arXiv:2109.05965.

- [8] *New estimates for exponential sums over multiplicative subgroups and intervals in prime fields*, (with D. Di Benedetto, M. Z. Garaev, V. C. Garcia, I. E. Shparlinski, and C. A. Trujillo). Journal of Number Theory 215, 261-274 (2020). Open access: arXiv:2003.06165.
- [9] *On sets with small sumset and m -sum-free sets in $\mathbb{Z}/p\mathbb{Z}$* , (with P. Candela and D. J. Grynkiewicz). Bulletin de la Société Mathématique de France 149 (1), 155-177 (2020). Open access: arXiv:1909.07967.
- [10] *A step towards the $3k - 4$ conjecture in $\mathbb{Z}/p\mathbb{Z}$ and an application to m -sum-free sets*, (with P. Candela and D. J. Grynkiewicz). Acta Mathematica Universitatis Comenianae (conference proceedings), Volume 88, (2019), Issue 3, pp. 521-525 (2019). Open access: arXiv:1909.07967.
- [11] *A note on the Bilinear Bogolyubov Theorem: Transverse and bilinear sets*. With P.-Y. Bienvenu and A. D. Martínez. Proceedings of the American Mathematical Society 148 (1), 23-31 (2020). Open access: arXiv:1811.09853.
- [12] *On nilspace systems and their morphisms.*, (with P. Candela and B. Szegedy). Ergodic Theory and Dynamical Systems, Volume 40, Issue 11, (2020) , pp. 3015 – 3029 . Open access: arXiv:1807.11510.
- [13] *A Plünnecke-Ruzsa inequality in compact abelian groups.*, (with P. Candela and A. de Roton). Revista Matemática Iberoamericana 35 (7), 2169-2186 (2019). Open access: arXiv:1712.07615.

Preprints

- [14] *MLPs at the EOC: Concentration of the NTK*, (with D. Terjék). Open access: arXiv:2501.14724.
- [15] *MLPs at the EOC: Spectrum of the NTK*, (with D. Terjék). Open access: arXiv:2501.13225.
- [16] *Spectral algorithms in higher-order Fourier analysis*, (with P. Candela and B. Szegedy). Open access: arXiv:2501.12287.
- [17] *On the inverse theorem for Gowers norms in abelian groups of bounded torsion*, (with P. Candela and B. Szegedy). Open access: arXiv:2311.13899.
- [18] *Free nilspaces, double-coset nilspaces, and Gowers norms*, (with P. Candela and B. Szegedy). Open access: arXiv:2305.11233.
- [19] *A framework for overparameterized learning*, (with D. Terjék). Open access: arXiv:2205.13507.

Contributed talks at conferences and workshops

- 07/2024 *A short proof of an inverse theorem in bounded torsion groups* at Discrete Mathematics Days 2024, Alcalá de Henares, Spain.
- 11/2023 *On measure-preserving $\mathbb{F}_p^\omega$ systems of order k* at the 11th workshop Operator Theoretic Aspects of Ergodic Theory, Wuppertal, Germany.
- 06/2023 *Free nilspaces, double-coset nilspaces, and Gowers norms* at Nilpotent structures in topological dynamics, ergodic theory and combinatorics, Będlewo Conference Center, Poland.
- 05/2022 *On affine-exchangeable measures and nilspaces* at the workshop Graphs, Groups, Stochastic Processes, Alfréd Rényi Institute of Mathematics, Budapest, Hungary.
- 09/2021 *A Refinement of Cauchy-Schwarz Complexity, with Applications*, at EuroComb 2021, online (Barcelona, Spain).
- 08/2019 *The $3k - 4$ theorem in $\mathbb{Z}/p\mathbb{Z}$ and an application to m -sum-free sets* at EuroComb 2019, Univerzita Komenského, Bratislava, Slovakia.

12/2017	<i>La desigualdad de Plünnecke-Ruzsa en grupos abelianos compactos</i> at the LXVI Reunión de Comunicaciones Científicas of RSME and UMA, Buenos Aires, Argentina.
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Invited talks at seminars

11/2024	<i>An algorithm for higher-order Fourier analysis</i> at Séminaire de Théorie des Nombres de Nancy-Metz, Institut Élie Cartan de Lorraine, France.
06/2024	<i>On measure-preserving $\mathbb{F}_p^\omega$-systems of order k.</i> at DAGGER seminar, University of Warwick, UK.
09/2022	<i>The Polyak-Łojasiewicz condition and overparameterized learning</i> (with D. Terjék) at Euler Seminar, Université catholique de Louvain, Louvain, Belgium.
03/2019	<i>Conjuntos grandes que se parecen a progresiones en grupos cíclicos de orden primo</i> (Large sets that resemble arithmetic progressions in cyclic groups of prime order), at Junior Seminar, Autonomous University of Madrid, Spain.
06/2018	<i>A Plünnecke-Ruzsa inequality in compact abelian groups</i> at UAM-ICMAT number theory seminar, ICMAT, Madrid, Spain.
03/2017	<i>El método polinomial en combinatoria aditiva</i> (the polynomial method in additive combinatorics) at the Junior Seminar, Autonomous University of Madrid, Spain.

Other events attended

07/2023	200 Years of Trinity Combinatorics, Cambridge, UK.
06/2022	Graphs, Groups, Stochastic processes summer school, Budapest, Hungary.
08/2019	Number theory in the Americas. Casa Matemática Oaxaca, Oaxaca, Mexico.
09/2018	Arithmetic Ramsey Theory, University of Manchester, Manchester, UK.
09/2017	The music of Numbers. Conference in honor of Javier Cilleruelo. ICMAT, Madrid, Spain.
05/2017	Graduate course on Interactions of harmonic analysis, combinatorics and number theory. BGSMATH, Barcelona, Spain.

Events organized

06/2025	(Expected) New Trends in Arithmetic Combinatorics and related Fields, a BIRS-IMAG workshop. Coorganized with: P. Candela (lead organizer), A. de Roton, H. Helfgott, A. Sedunova, and O. Serra. Check our website: https://birs-imag-ntac.github.io/ .
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Projects in which I have participated and role

03/2022 – 12/2025	<i>Artificial Intelligence National Laboratory (MILAB, RRF-2.3.1-21-2022-00004)</i> , NKFIH, Hungary. It partially funds the Research Fellow position at Alfréd Rényi institute of Mathematics. I work in the mathematical foundations of AI.
09/2021 – 08/2024	<i>Harmonic analysis, combinatorics and arithmetic</i> (PID2020-113350GB-I00) funded by MICINN, Spain. I am part of the working team.
11/2020 – 11/2022	<i>Momentum (Lendület) 30003</i> , NKFIH, Hungary. It partially funds the Research Fellow position at Alfréd Rényi institute of Mathematics. I work in HOF and in the mathematical foundations of AI.

11/2020 – 03/2025	“Élvonal” KKP 133921 grant, NKFIH, Hungary. It partially funds the Research Fellow position at Alfréd Rényi institute of Mathematics. I work in HOF and deep learning in this project.
01/2018 – 12/2020	<i>Arithmetic and harmonic analysis</i> , (MTM2017-83496-P) MINECO, Spain. I am part of the working team.

Research Fellowships/Prizes/Grants

15/07/2021	Extraordinary award to the best theses presented at the Autonomous University of Madrid in the period 01/10/2019 to 30/09/2020. Awarded by the Autonomous University of Madrid.
10/02/2017	La Caixa Severo Ochoa scholarship awarded by Fundación la Caixa (private company). Ph.D.’s funding. It included a monthly stipend, an amount for travels and other expenses, and all labour/tax costs.
27/06/2015	X Mutua Madrileña Posgraduate Scholarship awarded by Fundación Mutua Madrileña (private company). M.A.St.’s funding.
18/06/2014	JAE-intro scholarship for introduction to research awarded by CSIC, Ministry of Economy and Competitiveness, Spain.
03/12/2012	Madrid Excellence Scholarship awarded by the Autonomous Community of Madrid, Spain.
14/09/2011	Madrid Excellence Scholarship awarded by the Autonomous Community of Madrid, Spain.
26/01/2010	Madrid Excellence Scholarship awarded by the Autonomous Community of Madrid, Spain.

Accreditations

12/01/2022	<i>Profesor ayudante doctor</i> (assistant professor) granted by the Ministry of Universities of Spain.
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Graduate teaching

08/2021	Instructor at AGRA IV (Arithmetic, Groups and Analysis), online (Trieste, Italy).
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Undergraduate teaching

02/2024 – 05/2024	Professor (jointly with P. Zsámboi) of <i>Introduction to Deep Learning</i> at the Budapest Semesters in Mathematics (BSM), Spring Semester 2024 (28 hours taught). Note: This was the first time this course was taught at BSM. Thus, Zsámboi and I had to prepare it from scratch.
09/2019 – 11/2019	Assistant professor of <i>Mathematics I</i> for the B.Sc. in Chemical engineering at the Autonomous University of Madrid (6 hours taught).
09/2018 – 01/2019	Assistant professor of <i>Mathematics</i> for the B.Sc. in Environmental Sciences at the Autonomous University of Madrid (15 hours taught).
09/2017 – 01/2018	Assistant professor of <i>Algebra I</i> for the B.Sc. in Physics at the Autonomous University of Madrid (15 hours taught).

Service

Referee for	TEMat, <i>Analysis and PDE</i> , <i>AISTATS 2022</i> , <i>AISTATS 2024</i> and <i>The Ramanujan Journal</i> .
Reviewer for	<i>Zentralblatt MATH (zbMath)</i> .

Other outreach activities

22/04/2024	<i>Recent trends III: Subsets of the integers without three term arithmetic progressions</i> , an outreach article for the Discrete and Algorithmic Mathematics (DAM) network. Available at: https://dam-network.github.io/ .
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Languages

Human:	Spanish (mother tongue) English (fluent, used in professional contexts) French (basic).
Programming:	Python (including Pytorch), C, Java, SQL, Matlab, and R.